

Solution Requirements

Date	10 Jul2026
Team ID	XXXXXX
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>High</i> The system validates applicant information before processing loan eligibility prediction.	High
2	Authorization levels	The system provides loan eligibility prediction, applicant profile, prediction history, and report generation for authorized system usage.	Medium

3	External interfaces	<i>[Connects with the trained ML model to predict loan eligibility and uses local storage/database to save applicant records and prediction history.]</i>	High
4	Transactions processing	Processes applicant information, validates input data, generates loan eligibility predictions, and stores prediction results securely.	High
5	Reporting	<i>Generates prediction reports, applicant profiles, loan history, eligibility analysis, and downloadable PDF reports.</i>	High
6	RBusiness Rules	<i>Loan eligibility is determined based on applicant details, ML prediction score, and predefined eligibility criteria before generating the final recommendation.</i>	High
7	Compliance to laws or regulations	<i>Protects applicant information by ensuring secure data handling, maintaining privacy, and restricting unauthorized access to prediction records.</i>	Medium
8	Other	The system maintains applicant profiles, prediction history, and generates downloadable PDF reports for future reference.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	Category	Description	Acceptance Criteria
1	Performance & Speed	The system should generate loan eligibility predictions quickly with minimal response time.	Prediction result generated in less than 3 seconds .
2	Scalability	The system should efficiently handle multiple applicant records and prediction history.	Supports 5,000+ applicant records without performance degradation.
3	Security & Data Privacy	Applicant information should be securely stored and protected from unauthorized access.	Applicant data is stored securely with restricted access .
4	Reliability & Availability	The system should provide consistent prediction results and remain available during normal operation.	System availability of 99% with reliable prediction functionality.